1. $z = (2)(3) + (-1) = 5$

2. $z = (1.5)(1) + (-0.5) = 1$; $\hat{y} = \sigma(z) = \frac{1}{1+e^{-1}} = 0.73$

3. $z = (3)(-2) + (1) = -5$; $a = \text{ReLU}(z) = 0$

4. $L = \frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}(4-1)^2 = 9/2$

5. $L = \frac{1}{2}(\hat{y} - y)^2 \Rightarrow \frac{\partial L}{\partial \hat{y}} = (\hat{y} - y)(1) = 6 - 10 = -4$

6. $\frac{\partial L}{\partial w} = \frac{\partial}{\partial w}\left(\frac{1}{2}(\hat{y} - y)^2\right) = \frac{\partial}{\partial w}\left(\frac{1}{2}(wx + b - y)^2\right) = (wx + b - y)(x) = -6$

$\frac{\partial L}{\partial b} = \frac{\partial}{\partial b}\left(\frac{1}{2}(\hat{y} - y)^2\right) = \frac{\partial}{\partial b}\left(\frac{1}{2}(wx + b - y)^2\right) = (wx + b - y)(1) = -3$

7. $w_{new} = w_{old} - \eta \frac{\partial L}{\partial w} = 1 - 0.1(-6) = 1.6$

$b_{new} = b_{old} - \eta \frac{\partial L}{\partial b} = 0 - 0.1(-3) = 0.3$

8.

1) $\hat{y}_0 = w_0 x + b_0 = 0(1) + 0 = 0$; $\frac{\partial L}{\partial w} = -2$; $\frac{\partial L}{\partial b} = -2$

$w_1 = w_0 - \eta \frac{\partial L}{\partial w} = 0 - (0.5)(-2) = 1$; $b_1 = b_0 - \eta \frac{\partial L}{\partial b} = 0 - (0.5)(-2) = 1$

2) $\hat{y}_1 = w_1 x + b_1 = 1(1) + 1 = 2$; $\frac{\partial L}{\partial w} = 0$; $\frac{\partial L}{\partial b} = 0$

$w_2 = w_1 - \eta \frac{\partial L}{\partial w} = 1 - (0.5)(0) = 1$; $b_2 = b_1 - \eta \frac{\partial L}{\partial b} = 1 - (0.5)(0) = 1$